Noise and the fractal convergence boundary of gradient descent: first experiments

Arav Rawat · October 2026 · Follow-up to Liang & Montúfar [1] · Code and data: github.com/lnxv/fractal-convergence-noise

Summary. I first reproduced the deterministic results of [1]: the convergence region of Figure 3, the box-counting dimension 1.249 of the projected boundary, and the critical step size of Theorem 1. Adding noise (random step size, SGD on a two-sample finite sum, and isotropic Gaussian noise) turns the fractal boundary into a band of uncertain outcomes whose area scales as a power law in the noise level, while the location of the boundary barely moves. Noise affects *which* minimizer is selected more strongly than *whether* GD converges. At larger noise the behaviour changes qualitatively and differently for each noise model.

Question

Gradient descent (GD) with near-critical step size on regularized scalar factorization has a fractal convergence boundary [1]. Does a small amount of stochasticity preserve this boundary, blur it, or change it qualitatively, and what happens as the noise is increased further?

Step 1: recovering the deterministic phenomenon

Setting: $L(u,v) = \tfrac{1}{2}(uv - y)^2 + \tfrac{1}{2}\lambda(u^2 + v^2)$, $y = 0.5$, $\lambda = 0.2$, $\eta = 1$ (Figure 3 of [1]). (a) On a 401×401 grid over $[-4,4]^2$, GD reproduces the convergence region of [1, Fig. 3 middle]. (b) Iterating the planar map of [1, Prop. 3] on a 2000×2000 grid in $(uv,\ u^2 + v^2)$ and box-counting the boundary of the convergence region gives **D = 1.249 (R^2 = 0.9966)**, the value reported in [1]. (c) For $\lambda = 0$ and $y = 1$, GD with step size 3% (or 10%) below the critical step size of [1, Thm. 1] converged from all 20,000 random initializations, and 3% (or 10%) above it from none. (d) As an independent check I measured the uncertainty exponent of Grebogi, McDonald, Ott and Yorke [2]: the fraction of initial conditions whose outcome changes under a perturbation of size ε scales as ε^{2-D}. In the two windows used below this gives D ≈ 1.26 for the convergence boundary, and D ≈ 1.28–1.29 when the boundary between the two minimizers' basins is included.

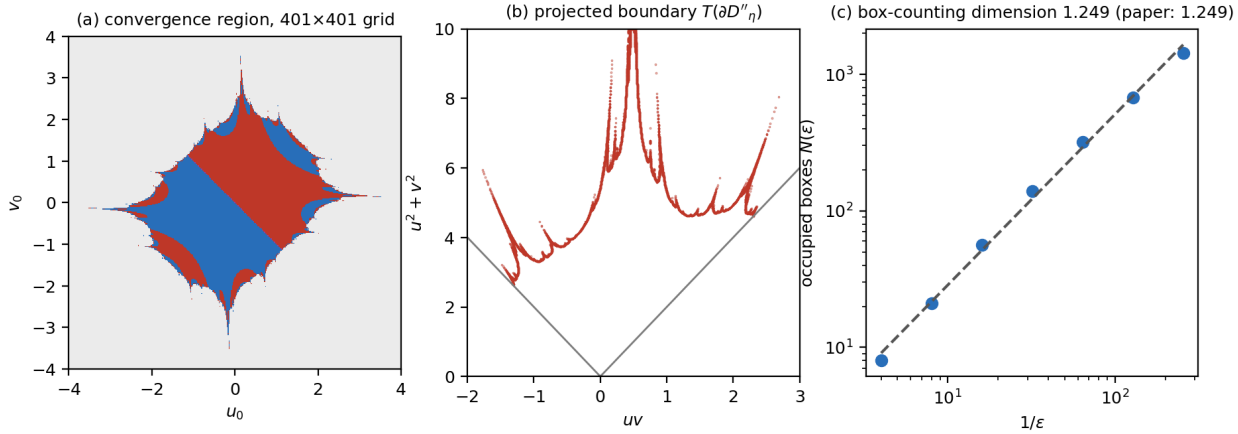


Figure 1. (a) Deterministic GD: red/blue converge to the two minimizers, gray escapes. (b) Projected boundary. (c) Box counting of (b).

Step 2: introducing stochasticity

Three noise models with level σ: **step**, a random step size $\eta_t = \eta(1 + \sigma\xi_t)$, $\xi_t \sim U(-1,1)$, under which the minimizers remain fixed points; **SGD**, batch-size-one SGD on the finite sum of two data points $y \pm \sigma$, which has the same minimizers as the deterministic loss and state-dependent gradient noise; and **additive** isotropic Gaussian gradient noise of standard deviation σ. With noise each initialization has a *probability* of each outcome. One noisy run per pixel is misleading: independent draws at neighbouring pixels produce speckle that looks like a rougher boundary (Fig. 4, right; interface fraction 0.059 versus 0.016 for a single shared noise path, which stays as sharp as deterministic GD). Following the probability-basin approach of Nieto, Seoane and Sanjuán [3], I estimate outcome probabilities from 96

independent runs per initialization (horizon 200 steps) on 101×101 grids in two windows near the boundary, W1 = [2, 2.7]×[−0.3, 0.3] and W2 = [1.7, 2.3]×[0.5, 1.1]. Two quantities are tracked: the fraction of initializations with 0.1 < P(no escape) < 0.9, and the fraction that usually survives but whose selected minimizer is uncertain (0.1 < P(+ minimizer | survive) < 0.9).

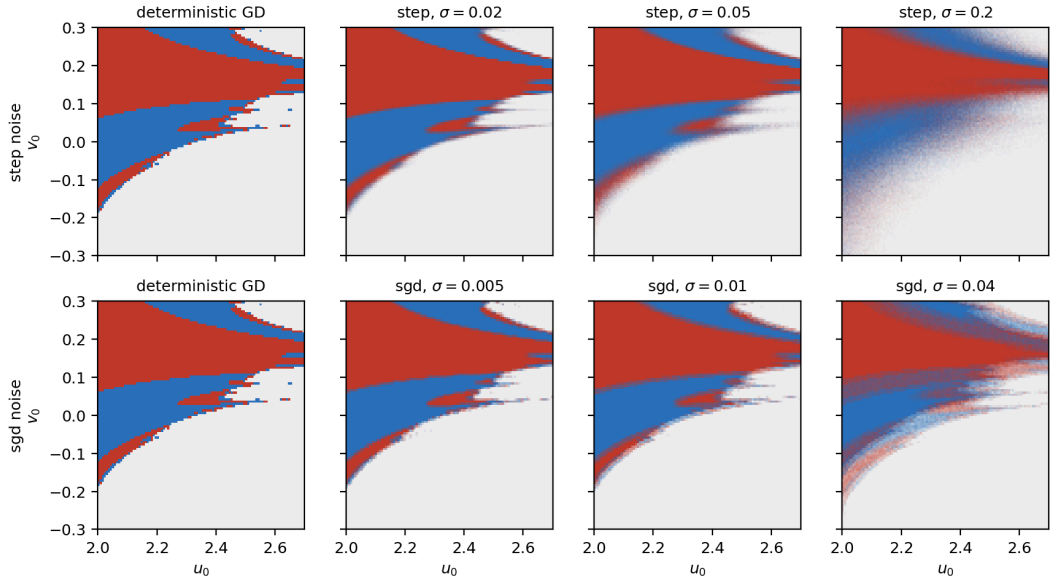


Figure 2. Outcome probabilities in W1 for step noise (top) and SGD (bottom); red and blue intensity give the probabilities of the two minimizers, gray of escape.

Findings near the boundary

- **Small noise blurs the boundary rather than preserving it.** Both the convergence boundary and the interface between the two basins become probabilistic bands (Fig. 2). Thin filaments wash out first: the p = ½ contour is 20–40% smoother than the deterministic boundary.
- **The boundary barely moves.** The majority outcome differs from deterministic GD on 0.1–3.9% of each window, and the probability of not escaping changes by at most 0.014.
- **Both uncertain fractions follow power laws in σ** (Fig. 3, $R^2 \geq 0.998$). The escape band is stable under 2× grid refinement (0.134 vs 0.135 at step σ = 0.1) and under doubling the horizon (0.136).
- **Selection is more fragile than convergence.** Exponents for minimizer selection (0.63–0.72) are at or below the deterministic value 2 − D, while those for escape (0.73–0.85) are mostly above it.

Exponent of the uncertain fraction	**whether GD escapes**		**which minimizer is selected**	
	W1	**W2**	**W1**	**W2**
step noise	0.85 ± 0.01	0.73 ± 0.01	0.64 ± 0.01	0.63 ± 0.01
SGD	0.80 ± 0.02	0.82 ± 0.02	0.69 ± 0.01	0.65 ± 0.01
additive noise	0.83 ± 0.01	0.85 ± 0.01	0.72 ± 0.002	0.65 ± 0.02
deterministic, perturbation ε (2 − D)	0.74 ± 0.01	0.74 ± 0.005	0.72 ± 0.01	0.71 ± 0.01

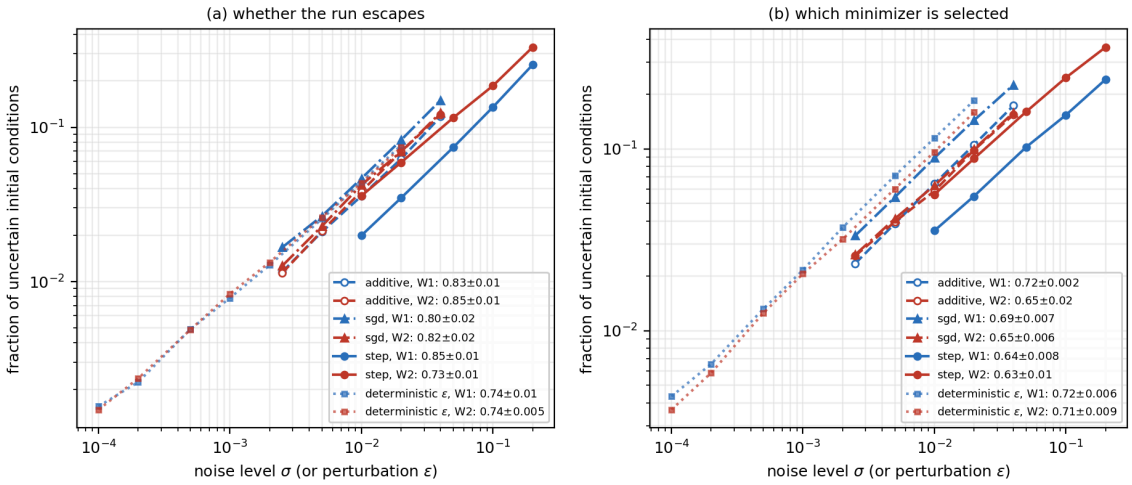


Figure 3. Fraction of uncertain initial conditions versus σ (noise) or ε (deterministic GD), log–log axes.

Step 3: increasing the noise further

Over the whole domain $[-4,4]^2$ (161×161 grid, 32 runs per initialization) each noise model eventually changes the picture qualitatively (Figs. 4–5):

- **Step noise enlarges the convergence region.** The probability of not escaping grows from 20.9% of the domain to 25.1% at σ = 0.8, and every surviving run still converges exactly. The minimizer is uncertain on up to about a third of the surviving region.
- **SGD has a threshold at σ = y − λ = 0.3.** Up to σ = 0.2, 99.7% of surviving runs end near a minimizer; at σ = 0.4 only 57% do. For σ > y − λ the sample loss with target $y - \sigma$ has its minimizer at the origin, so SGD is repeatedly pulled toward the saddle and the two basins lose their meaning.
- **Large isotropic noise randomizes the selection.** At σ = 0.2 the selected minimizer is uncertain over essentially the whole surviving region, and 14% of surviving runs end near neither minimizer.

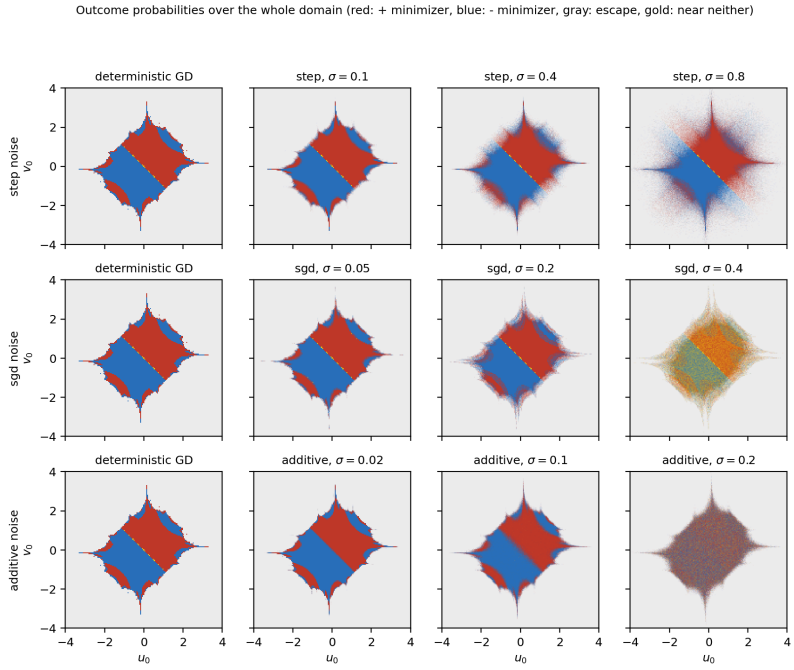


Figure 4. Outcome probabilities over $[-4,4]^2$ for increasing noise (rows: step, SGD, additive). Gold marks surviving runs that end near neither minimizer.

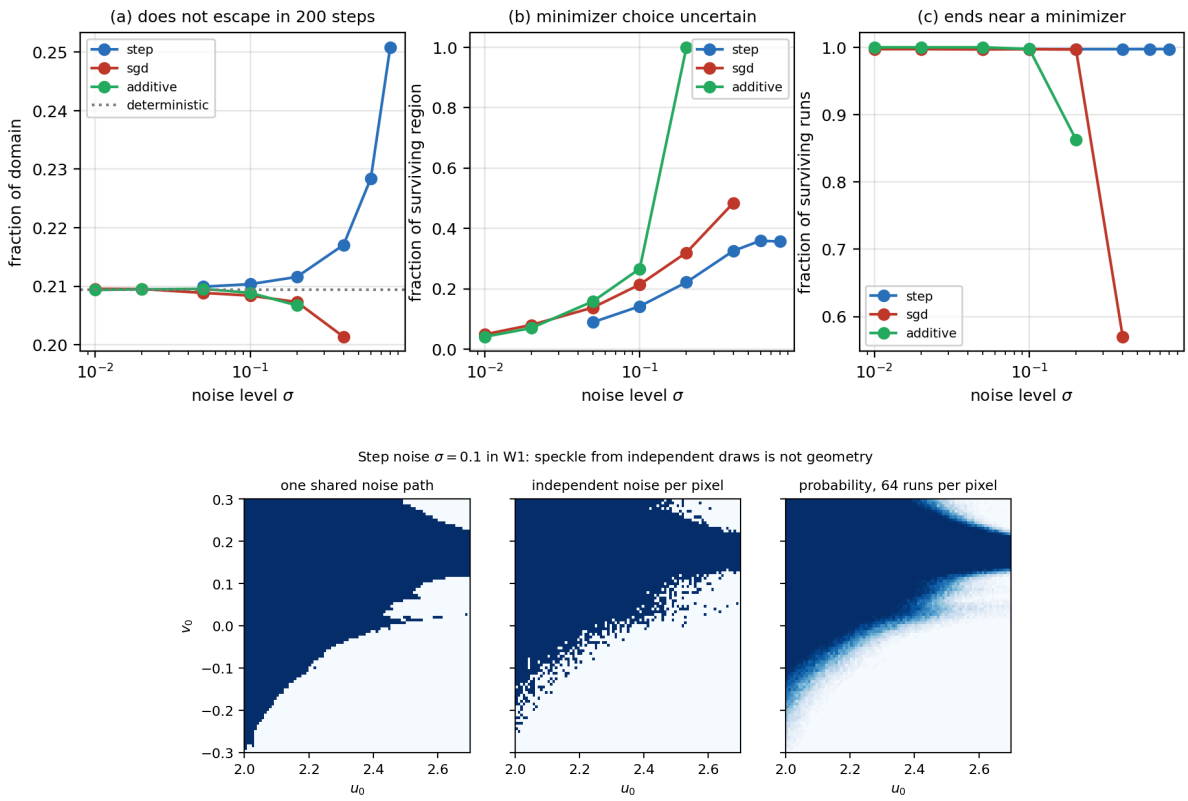


Figure 5. Top: whole-domain summaries. Bottom: step noise σ = 0.1 in W1 with one shared noise path, independent noise per pixel, and the probability map.

Interpretation and possible mathematical questions

If noise acted like a single random perturbation of the initialization of size proportional to σ, the exponents would equal 2 − D. The escape exponents are mostly larger and the selection exponents at or below that value, which suggests that noise injected at every step interacts differently with the two parts of the boundary. One possible explanation involves the folding of the GD map described in [1]. Natural questions: (i) can the noise exponents be derived from D and the expansion rate of the GD map on the boundary; (ii) why is selection more sensitive than convergence; (iii) can the enlargement of the convergence region under random step sizes be explained through the critical step size of Theorem 1. Sohl-Dickstein [4] observed that fractal trainability boundaries survive minibatch noise; if the minibatch sequence is fixed across the grid this corresponds to the shared-noise setting, where the boundary also stays sharp here.

Limitations

- Finite horizon: not escaping within 200 steps is not the same as asymptotic convergence; under persistent noise (SGD, additive) iterates fluctuate around a minimizer rather than converge.
- Two local windows; the 0.1/0.9 thresholds are a choice; 96 runs per point near the boundary and 32 over the whole domain.
- Scalar factorization (d = 1) only; matrix factorization and real minibatch training are not tested.

References

1. S. Liang and G. Montúfar. Gradient descent with large step sizes: chaos and fractal convergence region. ICLR 2026. arXiv:2509.25351.
2. C. Grebogi, S. W. McDonald, E. Ott and J. A. Yorke. Final state sensitivity: an obstruction to predictability. Phys. Lett. A 99 (1983) 415–418.
3. A. R. Nieto, J. M. Seoane and M. A. F. Sanjuán. Final state sensitivity in noisy chaotic scattering. Chaos, Solitons & Fractals 150 (2021) 111181.
4. J. Sohl-Dickstein. The boundary of neural network trainability is fractal. arXiv:2402.06184 (2024).